

Business Recommendations

Term-deposit subscription ranking for a controlled call-centre pilot

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Executive recommendation. Use unweighted Logistic Regression with all 12 pre-call features to rank already eligible customers from highest to lowest priority. Use the score as a ranking signal, not as a literal subscription probability or a rigid yes/no decision. Validate the ranked process in a controlled live pilot at equal calling effort before broader deployment. The recommended operating setup is summarized in Table 1.

Decision snapshot

Decision	Recommendation
Pilot model	Unweighted Logistic Regression
Inputs	All 12 pre-call predictors
Operational use	Rank eligible customers; call from the top of the list until capacity is reached
Challenger	Tuned unweighted HistGradientBoosting
Deployment status	Controlled pilot only
Probability use	Do not treat current scores as exact subscription probabilities

Table 1. Recommended pilot configuration and operating role.

1. Business Case

The historical data contain 40,000 completed marketing calls and 2,896 subscriptions, a conversion rate of 7.24%. Because 92.76% of outcomes are non-subscriptions, accuracy alone is not a useful business test: an always-no rule is already 92.76% accurate.

The useful question is therefore whether the model can place more future subscribers near the top of a limited call list. Logistic Regression is recommended for the pilot because the nonlinear challenger is stronger on shuffled historical folds, but Logistic Regression transfers better across later ordered campaign conditions and produces stronger period-local lift. The evidence supporting that choice is summarized in Table 2.

Evidence	Logistic Regression	HistGradientBoosting	Interpretation
Shuffled ROC-AUC	0.685	0.715	HGB is stronger in shuffled development validation
Ordered ROC-AUC	0.566 +/- 0.040	0.553 +/- 0.022	LR is more robust across later ordered conditions
5% call depth	Lift 1.85	Lift 1.31	LR concentrates more subscribers near the very top
20% call depth	Lift 1.40	Lift 1.27	LR retains the stronger operational ranking result

Table 2. Historical model comparison and operational ranking evidence.

The ordered test is a robustness proxy, not a definitive future-calendar validation. The source data contain month labels but no verified year, campaign ID or trustworthy timestamp.

2. Operational Value

The model ranks eligible customers by relative priority. A call depth of 5%, for example, means calling only the highest-ranked 5% of the eligible list. Lift compares the number of subscribers captured by that ranked list with the number expected under random selection at the same call volume. Historical subscriber capture and lift at the illustrative call depths are summarized in Table 3.

Call depth	Calls	Subscribers captured	Expected random	Lift
5%	1,528	228	123	1.85
10%	3,055	395	246	1.61
20%	6,108	689	492	1.40

Table 3. Historical subscriber capture and lift at illustrative call depths.

These historical call depths are operating examples, not fixed rules. The business should choose depth from current capacity and economics. The exploratory higher-conversion population contains 6,774 customers and 778 subscribers, with an 11.49% historical conversion rate versus 7.24% overall, but uncertainty analysis does not justify a separate model. Use the population for monitoring and hypothesis testing rather than as a hard targeting rule.

Current model scores should be interpreted as relative ranking scores. Ordered calibration is not reliable enough to convert the raw score directly into a claimed subscription probability or expected profit.

3. Pilot and Controls

The live test should compare Logistic-Regression-based prioritisation with the best available benchmark at the same calling effort. If a clear existing targeting process is defined, use it as the control. If no reliable process exists or it cannot be reconstructed, use random selection from the same eligible pool.

- Randomly assign comparable eligible customers to treatment and control groups.
- Treatment: rank customers with the selected Logistic Regression model and call from the top of the list.
- Control: use the existing targeting process, or random selection if no reliable process exists.
- Keep call volume and approximate staff calling time comparable between groups.
- Primary outcome: additional subscriptions at the same calling effort; also report subscriptions per 1,000 calls.
- Record calls, staff-hours, subscriptions, calling cost and the financial value of successful subscriptions.
- Select final call depth from the point at which the value of additional subscriptions no longer justifies additional calling effort. The principal pilot risks and controls are summarized in Table 4.

Risk	Control
Changing campaign conditions	Track live lift and response rate by campaign period; stop or reassess if performance deteriorates materially.
Uncalibrated scores	Use scores for ranking only until prospective recalibration is supported.
contact=unknown	Audit the operational meaning, pre-call availability and recoverability of this field.
Exploratory segment instability	Monitor the segment but do not deploy a separate model without stronger prospective evidence.

Table 4. Principal pilot risks and proposed controls.

4. Data and Next Steps

Better operational data may add more value than further tuning of the same historical feature set. Priority improvements are reliable campaign dates and IDs, previous contact attempts and outcomes, time since prior contact, clearer contact-channel information and richer CRM history.

- Prepare the controlled pilot and freeze eligibility, scoring and measurement rules before launch.
- Run the pilot and compare incremental subscriptions at equal calling effort.
- Measure cost per acquisition, contribution per successful subscription and realised lift.
- Audit contact-channel data quality, especially the large contact=unknown category.
- Recalibrate scores on prospective data before probability-based revenue or expected-value calculations.
- Expand deployment only if the pilot demonstrates repeatable incremental value under current operating conditions.